Muhammad Hammad Faisal

Software Engineer - Python, Backend & Automation

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years building Python and backend systems alongside production automation in TypeScript and Node.js. Proposed and built a FastAPI microservice for PDF data extraction (factory pattern) that was adopted across multiple projects, built Django back ends including a full Learning Management System, and wrote a Python ShadowSocks VPN client. Consolidated 23 service integrations under one architecture in two weeks and cut a 100+ file test suite's runtime from 8 hours to 2. Comfortable with PostgreSQL, Docker, CI/CD on GitHub Actions, GitLab CI and Jenkins, code review, and Agile/Scrum teams.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, Go, HTML5, CSS3

Backend & APIs: FastAPI, Django, Django Channels, REST APIs, WebSockets, Node.js, PostgreSQL, Firestore

Automation & Scraping: Playwright, Patchright, Puppeteer, Selenium, WebdriverIO, Appium, CAPTCHA solving (reCAPTCHA v2/v3, Amazon), rotating proxies

CI/CD & Cloud: GitHub Actions, GitLab CI, Jenkins, Git, Docker, Google Cloud (Cloud Functions, Cloud Run), Vercel, Supabase

Practices: Factory and abstraction patterns, State machines, Code review, Agile / Scrum

Tools: Jira, Postman, Supertest, Allure Reports, Mocha

PROFESSIONAL EXPERIENCE

Software Engineer **June 2024 - Present**

Arbisoft - Lahore, Pakistan | Client products: FixAlert (test automation) and Zuub (data pipelines)

- Proposed and architected a FastAPI microservice for PDF data extraction using the factory pattern, chosen over Node.js for Python's parsing ecosystem; adopted across multiple projects.
- Consolidated 23 separate payer integrations under a unified bot architecture to eliminate session conflicts, refactoring and merging shared modules within a 2-week deadline; received public recognition and compensatory leave from the Team Lead.
- Built a Python Outline-style VPN client (ShadowSocks) to unblock geo-restricted targets that macOS networking could not reach.
- Built and maintain Zuub's TypeScript automation bots (Puppeteer, Playwright, Patchright): login flows, CAPTCHA handling via 2Captcha, and extraction of data into a fixed interface schema.
- Own the credential lifecycle for every bot (validation, password update and reset) through a state-based rotation system managing thousands of accounts with zero bans or lockouts; verify extracted data is complete and correct rather than trusting run success.
- Refactored FixAlert's inherited 100+ file WebdriverIO suite to 60 files and cut full-suite runtime from 8 hours to 2 with parallel execution and a distributed Selenium Grid; wired CI/CD via GitHub Actions.
- Standardized TypeScript conventions and PR-review workflows across the team.

Jr. Software Engineer - Test Automation **May 2023 - June 2024**
Arbisoft - Lahore, Pakistan

- Designed a framework abstraction layer so the codebase depended on one interface rather than a third-party tool, enabling a full framework migration by rewriting only the adapter; reused across teams.
- Built a cross-platform automation framework using the factory pattern to unify Android, iOS and web configuration.
- Implemented a local Selenium Grid with Appium for parallel execution, cutting regression testing time by 40%.
- Collaborated with QA, dev and product teams on automation strategy; helped onboard the Zuub project and set up its automation infrastructure.

Freelance Software Engineer **April 2023 - December 2024**
Upwork - Remote

- Built scraping infrastructure with Patchright and custom anti-detection techniques for client data projects.
- Delivered multi-framework test automation across Playwright, WebdriverIO, Cypress and Selenium; set up CI/CD pipelines.

Jr. Software Engineer **August 2022 - May 2023**
Ginkgo Retail - Lahore, Pakistan

- Developed a Learning Management System end-to-end: Django back end, React front end, and deployment.
- Built and maintained web applications with Django and React across the full development lifecycle; contributed to CI/CD pipelines and code reviews.

KEY PROJECTS

ProxyPal - macOS desktop client for managing ShadowSocks connections, inspired by Outline. Python. github.com/M-Hammad-Faisal/ProxyPal

Stealth Scraper (Patchright) - Stealth scraping library with six anti-detection techniques. TypeScript, GitHub Actions CI. github.com/M-Hammad-Faisal/stealth-scraper-playwright

SwagLabs Automation Framework - 50 tests, Selenium and Playwright engines, pytest-xdist, Page Object Model, Allure, GitHub Actions. Python. github.com/M-Hammad-Faisal/Website-Tester

EDUCATION

Bachelor of Engineering - Computer Engineering, University of Engineering and Technology, Lahore. Oct 2019 - May 2023 | GPA 3.60 / 4.0

CERTIFICATIONS

Python for Everybody Specialization - University of Michigan / Coursera

ADDITIONAL

Open Source: 7 merged PRs across the WebdriverIO org. **Languages:** English (Professional), Urdu (Native).